100 Days of Python

Day 55 — Practice Assignment: Problem Solving

For every problem, use the six steps. Before you write any code, write down the inputs, the output and a short plan. Work out the answer for the sample by hand, then test your code on the sample and on at least one edge case.

1. **Find the highest mark in the list [45, 67, 89, 34, 90] without using max(). The answer for this sample is 90. Decide in advance what your function should do for a list with a single mark.**

   Hint: plan it as "assume the first mark is highest, then check each mark and update if a bigger one appears".

2. **Write a function that counts the vowels (a, e, i, o, u) in a word. "Programming" has 3 and "Python" has 1. Make sure your function also counts capital letters, so "AEIOU" gives 5, and test a word with no vowels such as "rhythm".**

   Hint: converting the word to lowercase first means you only need to check five letters.

3. **A shop adds a delivery charge of 40 to any bill under 500. Orders of 500 or more are free. Given a list of item prices, return the final bill. [200, 150, 100] should give 490, and [300, 250] should give 550. Decide what an empty cart should return, and test the boundary of exactly 500.**

   Hint: work out the total first, then decide whether the charge applies. Is 500 "under 500"?

4. **Find the second-highest distinct mark in a list. For [90, 90, 80, 70] the answer is 80, not 90. Decide what to return when there is only one distinct mark, such as [50].**

   Hint: step 1 matters here. Removing duplicates with set() and then sorting from highest to lowest makes the plan short.

5. **Write a function that checks whether a word is a palindrome, meaning it reads the same forwards and backwards. "madam" should give True and "python" should give False. Decide whether "Madam" should count, and test a single-letter word.**

   Hint: a word[::-1] slice gives the word reversed. Compare it with the original.